

# Surface Segregation of Zinc Oxide Nanoparticles in Polymer Films and Fibers: Implications for Functionalized Fabrics

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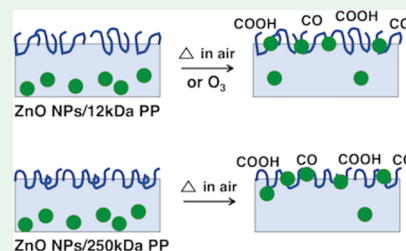
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**ABSTRACT:** Fibers and fabrics whose surfaces are decorated with metal oxide particles are finding important applications, including antimicrobial, filtration, and catalytic ones. In the present study, a very simple method has been developed to functionalize polypropylene (PP) surfaces, including those of fibers, with metal oxide nanoparticles (NPs). Extruded fibers and spin-coated films have been prepared from solid mixtures of PP and zinc oxide NPs using either 12 or 250 kDa PP (PP12k and PP250k, respectively). In the case of PP12k films containing 5 wt % ZnO, the segregation of the NPs to the polymer–air interface is detected by the emergence of Zn2p X-ray photoelectron spectroscopy (XPS) signal after either room temperature ozone exposure or annealing at 140 °C in an oxygen-containing atmosphere. For PP250k films, ozone exposure without heating does not induce surface segregation; however, it does occur upon annealing at 160 °C in an oxygen-containing atmosphere. Annealing in air causes the atomic percentages of Zn detected by XPS to increase from 0 to 0.22 and 0.34% for the PP12k and PP250K films, respectively. Cross-sectional scanning electron microscopy of a 3 wt % ZnO in PP12k fiber shows that NP migration to the surface does not deplete the bulk of ZnO, implying that the migration of NPs to the surface occurs from the near-surface region. Adsorption of a phosphonated dye, flavin mononucleotide, on surface-segregated films and fibers is confirmed by fluorescence microscopy, indicating that the ZnO NPs are available for surface reactions. XPS and Fourier transform infrared spectroscopy of PP exposed to ozone or heated in air demonstrate the presence of ketone and carboxylic acid functional groups. Heating of a ZnO surface-segregated PP12k film in ultrahigh vacuum to partially reverse oxidation also reverses surface segregation. It is postulated that surface segregation is enthalpically driven by the attraction of the ZnO NPs to the polar surface functional groups. This work may be useful for the preparation of fibers and fabrics for catalysis/photocatalysis applications.

**KEYWORDS:** melt extrusion, polypropylene, electron microscopy, surface segregation, photoelectron spectroscopy, zinc oxide



## INTRODUCTION

Metal oxide nanoparticles (MONPs) are technologically relevant,<sup>1–3</sup> and the functionalization of fibers with MONPs has various applications that include antimicrobial,<sup>4–6</sup> flame retardant,<sup>7</sup> air filtering,<sup>8</sup> and ultraviolet (UV) light-resistant<sup>9,10</sup> fabrics. Additionally, fibers and fabrics decorated with catalytic or photocatalytic MONPs may be useful for hazardous chemical degradation and environmental remediation.<sup>11–13</sup> Photoluminescent metal oxide-decorated fibers or fabrics may also be used for chemical sensing<sup>11</sup> and anti-counterfeit applications.<sup>14</sup>

Methods of functionalizing polymer surfaces, including those of fibers/fabrics, with metal oxide nanoparticles may be generally grouped into three classes. The first consists of pretreatment to increase NP adhesion, such as with plasma, ozone, or chromic acid.<sup>11,15</sup> These treatments lead to the formation of reactive groups (e.g., carboxylic acids) that can form covalent linkages to the NPs. The second is chemical grafting of NPs to the surface. For example, Zhang and colleagues<sup>16</sup> used a silane coupling agent to graft hematite

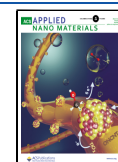
nanoparticles to polyethylene terephthalate fibers. The third class includes the growth of the nanoparticles directly onto the surface of the fiber using atomic layer or chemical vapor deposition.<sup>17</sup>

In an interesting study, Peter et al.<sup>18</sup> observed surface enrichment in iron oxide nanoparticles of an electrospun polyacrylonitrile (PAN) membrane when sodium dodecyl sulfate (SDS) was used in the electrospinning sol gel. The surface enrichment was believed to be due to SDS transport of the iron oxide nanoparticles to the surface driven by increased surface hydrophobicity and decreased surface free energy. The SDS could then be removed by washing in water.

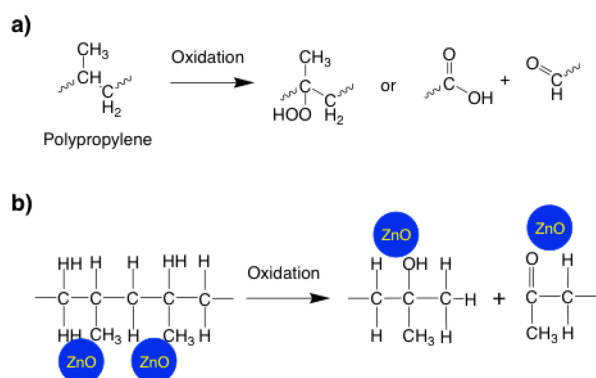
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In the present work, we have taken a different approach to induce surface segregation. A mixture of solid PP and ZnO NPs has been melt extruded to form fibers, or spin-coated into films, that contain embedded NPs. It is demonstrated that the oxidation of the surface of the fibers and films, resulting in the formation of ketone and carboxylic acid groups, causes ZnO NPs to surface segregate. Their availability for surface reactions is confirmed by the selective adsorption of a phosphonated fluorescent dye onto the NPs. The surface segregation mechanism has been investigated using both fibers and spin-coated films, and it is hypothesized that the process is enthalpically driven by the attraction of the ZnO NPs to the polar groups on the polymer surface. A schematic of the process is depicted in Figure 1.



**Figure 1.** (a) Examples of the types of products that result from the oxidation of polypropylene (PP) films by heating in air or ozone exposure. Other oxidation products may also be present, as discussed in ref 15. (b) A simplified schematic of the surface segregation process being investigated.

## EXPERIMENTAL SECTION

**Spin-Coated Polypropylene Films and Zinc Oxide/Polypropylene Films.** PP and ZnO/PP films were prepared with a Headway Research spin-coater (model 1-EC101D-R485) from commercially available PP and ZnO nanoparticles (Nanostructured & Amorphous Materials, Inc., diameter 20–40 nm, stock no. 5811HT). A few experiments were also performed with silica nanoparticles (Nanostructured & Amorphous Materials, Inc., diameter 20 nm, stock no. 4860MR). Two different isotactic PPs were used with molecular weights ( $M_w$ ) of ca. 12,000 Da (referred to as “PP12k”) and 250,000 Da (referred to as “PP250K”): Aldrich stock nos. 428116 and 427888, respectively. In the case of ZnO/PP films, 10 wt % of PP was suspended in hot *o*-xylene, and then 0.5 wt % of ZnO nanoparticles was added to the colloidal suspension with stirring. For example, 0.88 g of PP and 0.044 g of ZnO were added to 8.8 g (10 mL) of *o*-xylene. Brief sonication was periodically required to maintain the dispersion of the nanoparticles. Si wafers (ca.  $1 \times 1 \text{ cm}^2$ ) were used as substrates. The spin-coating suspension, Si wafers, and spin coater chuck were kept hot, using either a hot plate at a nominal temperature of 115 °C or a heat gun, until the coating process was finished to obtain uniform films. For spin-coating, 100  $\mu\text{L}$  of the hot suspension was dispensed on the center of the substrate at a rotational speed of 2500 rpm, with a rotational duration for 30 s. This procedure yielded dry films that were ca. 5 wt % in ZnO. Pure PP films were produced similarly without adding ZnO nanoparticles to the PP suspension. The spin-coating procedure gave a typical thickness, as determined by ellipsometry, of 800–900 nm.

**Melt-Extruded Polypropylene Fibers and Zinc Oxide/Polypropylene Fibers.** ZnO-incorporated single monofilament PP fiber samples were fabricated using a micro-compounder (manufactured by DACA, Inc.) coupled with a co-rotating twin screw extruder

at 100 rpm and a spinneret with a diameter of 0.75 mm. Before processing on the micro-compounder, the ZnO was dried overnight in a vacuum oven at 200 °C. When the barrel reached a stable operating temperature (160 °C for PP12k and 170 °C for PP250k), 3 g of PP containing 0.09 g of ZnO nanoparticles (3 wt %) was loaded into a hopper and injected into the barrel, which continued mixing for 3 min. As the polymer material flowed out of the spinneret, which was connected to the barrel outlet orifice, the fiber was collected onto a wider system composed of a Stedi-Speed Stirrer with a modified arm that secured a miniature bobbin. The winder system elongated and collected the material strand from the spinneret to produce monofilament fibers. Since PP12k lacked structural material strength to tolerate the elongation process on the winder, these fibers were processed without the spinneret and directly collected from the orifice.

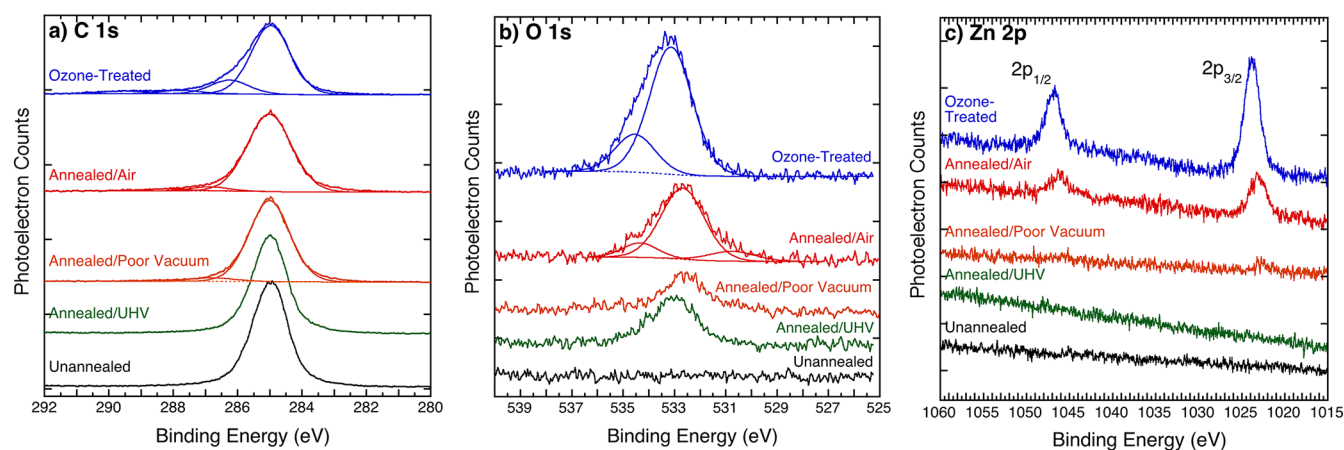
**Oxidation of Films and Fibers to Induce Surface Segregation.** Two different methods were investigated for inducing surface segregation: (1) annealing in ultrahigh vacuum (UHV), in poor vacuum (ca. 8 mbar), or in air and (2) oxidation by ozone treatment. Annealing samples in UHV was performed with a resistively heated sample stage in the X-ray photoelectron spectroscopy (XPS) instrument (ThermoVG Sigma Probe), which has a base pressure in the low  $10^{-9}$  mbar range. For the studies in poor vacuum and at atmospheric pressure, a vacuum oven (1415M, VWR Shel Lab) connected to a dry pump (Drytel 1025, Adixen) was used. For ozone exposure, spin-cast PP films or fibers were positioned in an Erlenmeyer flask and exposed to ozone admitted through a Teflon tube for 60 min at room temperature using an ozone generator (MKS, AX8407) with an output of 67–68% at an oxygen flow rate of ca.  $1 \text{ L min}^{-1}$ .

To explore the possibility that annealing may increase chain mobility and permit diffusion to the surface, PP12k and PP250k films containing 5 wt % ZnO were annealed at 140 and 160 °C, respectively, for 24 h in UHV, in poor vacuum (ca. 8 mbar), and in air. These temperatures were chosen because they were below the melting points of pure PP12k and PP250K, which were measured by differential scanning calorimetry as 158.0 and 166.5 °C, respectively.

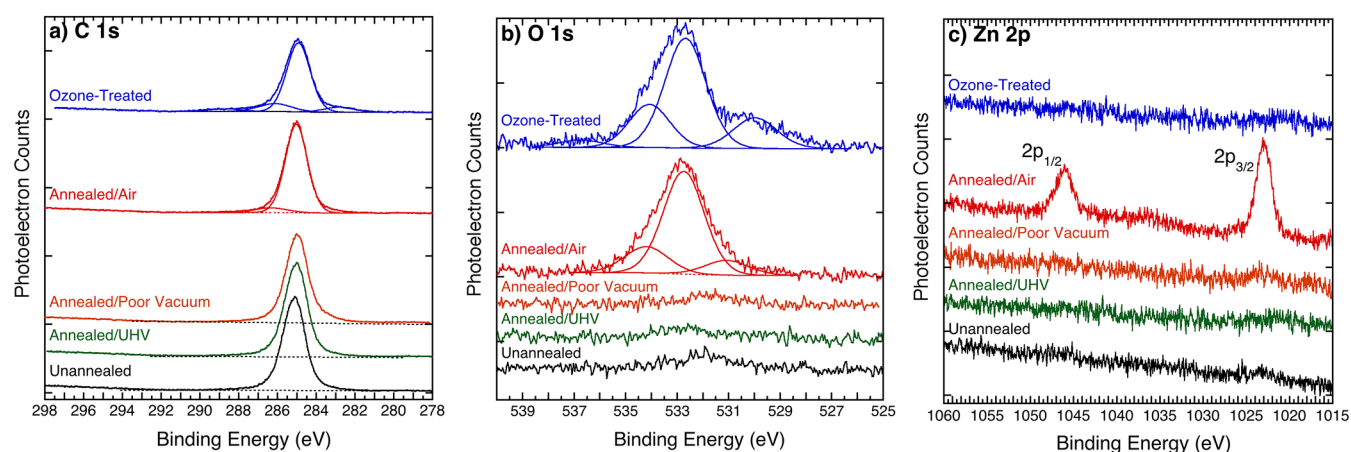
**Characterization of Fibers and Films.** Field-emission scanning electron microscopy (FE-SEM) and energy dispersive spectroscopy (EDS) were performed using a JEOL JSM 7401F instrument to obtain images of the cross-sectional areas of unannealed and annealed fibers. Cross-sectioning of PP fibers was achieved by microtoming after curing them at room temperature in an epoxy block consisting of a resin and hardener in a 6:1 v/v ratio. Subsequently, the microtomed block was sputter coated with gold prior to SEM analysis to eliminate electron charge buildup. EDS mapping images were obtained using an accelerating voltage of 20 kV, with a dwell time of 200 s.

XPS surface analysis of the spin-coated PP films was performed in the instrument discussed earlier using Al  $K\alpha$  X-rays ( $h\nu = 1486.6 \text{ eV}$ ), with an analyzer pass energy of 20 eV, take-off angle of ca. 37°, and energy step size of 0.05 eV. The PP films on silicon wafers were attached to the sample stage using a vacuum-compatible, double-sided adhesive copper tape. The spectra were corrected for surface charging, which occurs due to the incomplete compensation of ejected photoelectrons, by shifting the entire spectral region by the energy necessary to align the C1s peak to 285.0 eV, which is the generally accepted value for PP.<sup>19</sup> For quantification, the areas of the peaks were divided by the appropriate sensitivity factors. For peak fitting, a Shirley background and Gaussian (70%)–Lorentzian (30%) simulated peaks were used. In some cases, samples were heated in UHV, and this was accomplished by using a heatable sample stage, with the silicon wafer held to the ceramic heater, which contained a thermocouple, via clips on the edge of the sample. In all cases, the annealing duration was 24 h. The spin-coated film XPS experiments presented in this paper were performed on multiple samples (at least four) to confirm that the results shown are typical and reproducible.

A Nicolet iS10 FTIR was employed for attenuated total reflection Fourier transform infrared (ATR-FTIR) measurements to determine the functional groups present on heated and ozone-treated PP. In the case of the untreated and heated PP, outer layers of several PP pellets



**Figure 2.** Al  $K\alpha$  XPS of the (a) C1s, (b) O1s, and (c) Zn2p regions of spin-coated films of ZnO/PP12k on silicon wafers treated in different ways. The annealed samples were heated at 140 °C for 24 h under the conditions noted near the curve. The "ozone-treated" sample was exposed to ozone gas for 1 h. Peak fits are included for spectra that have more than one significant component.



**Figure 3.** Al  $K\alpha$  XPS of the (a) C1s, (b) O1s, and (c) Zn2p regions of spin-coated films of ZnO/PP250k on silicon wafers treated in different ways. The annealed samples were heated at 160 °C for 24 h under the conditions noted near the curve. The "ozone-treated" sample was exposed to ozone gas for 1 h. Peak fits are included for spectra that have more than one significant component.

were scraped off, and ATR was performed on the scrapings. For the ozone-treated PP sample, a pellet was ground into powder using a mortar/pestle, exposed to ozone for 1 h, and then analyzed.

**Assessment of Surface Activity.** To determine whether the surface-segregated ZnO NPs were available for adsorption, a 1.0 mM solution of flavin mononucleotide (FMN) (a.k.a. riboflavin-5'-phosphate) was freshly prepared in deionized water. The film and fibers samples were immersed in the FMN solution for 30 min, with room light excluded, followed by copiously rinsing them several times with deionized water to remove the excess dye. In the case of films, fluorescence spectra were obtained with a Spex Fluorolog 2 fluorometer operating in front-face reflection mode using an excitation wavelength of 440 nm with 3 mm excitation and emission slit widths. This excitation wavelength was chosen to excite the dye molecules but minimize the fluorescence excitation of the ZnO NPs. In the case of fibers, confocal laser fluorescence microscopy using an Olympus LSCM FV300 instrument was performed with an argon ion laser excitation source operating at 458 nm and a 510–570 nm long-pass filter. After immersion of fibers in the FMN solution, they were placed on a microscope slide, and a cover glass was attached using clear nail polish.

## RESULTS AND DISCUSSION

**Spin-Coated PP Films.** One possible method to induce surface segregation of metal oxide NPs embedded in PP films

is annealing. The increased mobility of the polymer chains lowers diffusion barriers and may enable migration to the surface, assuming it is thermodynamically favorable. XPS was performed on annealed films to determine whether ZnO NPs had diffused to the surface. Figures 2 and 3 show XPS of the C1s, O1s, and Zn2p regions of these samples and unannealed controls, and the oxygen and Zn percentages (based on carbon) are displayed in Table 1. For both the low and high molecular weight PP films, the unheated samples and the ones heated in UHV (in which negligible oxygen is present) show very small Zn2p signals. Heating in poor vacuum and in air leads to surface oxidation and emergence of peaks at 1022.7 and 1045.8 eV, corresponding to the spin-orbit coupled Zn2p<sub>3/2</sub> and Zn2p<sub>1/2</sub> peaks, indicative of the surface segregation of the NPs. After annealing in air, the atomic percentage of Zn is 0.22 and 0.34% for the PP12k and PP250K samples, respectively. The oxygen percentage also increases from 0 to 7.1 and 6.6% for the annealed PP12k and PP250k samples, respectively. While a small fraction of the O1s signal originates from ZnO (because of the 1:1 stoichiometry), most of it arises from the oxidation of the polymer surface.

The C1s spectra in Figures 2 and 3 show a similar behavior and indicate that ozone exposure and heating in air are

**Table 1. Atomic Concentration as Percent of Carbon, as Determined by XPS, of Spin-Coated PP Films Containing 5 wt % of ZnO NPs after Various Treatments<sup>a</sup>**

| sample                                | oxygen (%) | zinc (%) |
|---------------------------------------|------------|----------|
| untreated ZnO/PP12k                   | 0          | 0        |
| ZnO/PP12k + ozone                     | 11         | 0.66     |
| ZnO/PP12k + ozone, then 140 °C in UHV | 2.9        | 0.13     |
| ZnO/PP12k + annealed in UHV           | 1.6        | 0        |
| ZnO/PP12k + annealed in poor vacuum   | 2.9        | 0.04     |
| ZnO/PP12k + annealed in air           | 7.1        | 0.22     |
| Untreated ZnO/PP250k                  | 0.86       | 0.024    |
| ZnO/PP250k + ozone                    | 9.7        | 0.050    |
| ZnO/PP250k + annealed in UHV          | 0.60       | 0.032    |
| ZnO/PP250k + annealed in poor vacuum  | 0.68       | 0.034    |
| ZnO/PP250k + annealed in air          | 6.6        | 0.34     |

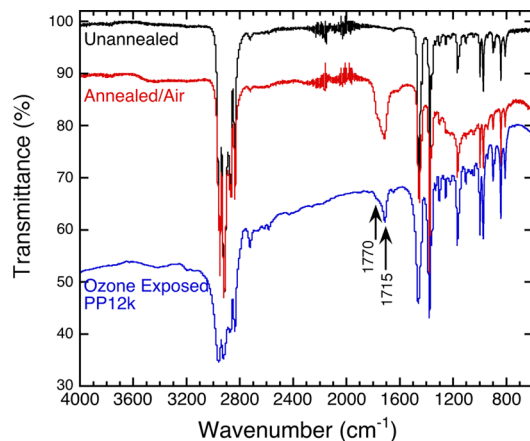
<sup>a</sup>The C1s, O1s, and Zn2p core levels were used for these analyses. The experimental error in the atomic percentages, determined from multiple sample analyses, is ca. 10%.

especially effective at oxidizing the surface of the PP films. While XPS is a surface-sensitive technique, it still detects atoms from ca. 100 Å deep. Because only the top layer of the surface is oxidized by heating in poor vacuum or air, the photoelectrons from the top layer contribute a small amount to the overall C1s signal. Exposure to ozone is more aggressive though, likely oxidizing more than just the top polymer layer. It should be noted that because a corona discharge type ozone generator was used, no heat or ultraviolet light exposure of the sample occurs during ozone exposure.

For the samples that were treated with ozone or annealed in air, a tail to the high binding energy side of the main (aliphatic) C1s peak at 285.0 eV is obvious. In the case of the ozone-treated PP12k sample, there are three additional components at 286.3, 287.8, and 289.5 eV, which may be assigned to C–O–C and/or C–OH, C=O, and O=C–OH groups, respectively.<sup>19</sup> The air-annealed PP12k sample shows components at 285.0, 286.8, and 288.2 eV, with the latter two likely due to C–O–C/C–OH and C=O groups, respectively. The C1s spectra for PP250k (Figure 3a) exhibit a similar behavior.

More dramatic changes occur in the O1s spectra since oxygen is not in the bulk of the PP sample and does not overwhelm the surface contribution. As shown in Figures 2 and 3, the O1s peak at 532.8–533.00 eV increases in intensity as the films are heated in oxygen-containing atmospheres and upon ozone treatment; this peak is ascribed to oxygen atoms in C–OH and/or C=O groups.<sup>15</sup> Ozone treatment causes the emergence of a second component at ca. 534.3 eV that is likely due to carboxylic acid groups. For the ozone-treated PP250K sample, an additional component at ca. 536–537 eV is required to fit the experimental spectrum (Figure 3b), and this is likely due to oxygen atoms in ester, perester, or hydroperoxide surface species.<sup>15</sup> A small component at ca. 530.2 eV is required for some of the spectra; the origin of this peak is uncertain, but it is possibly an artifact of inhomogeneous surface charging.

To further study the nature of the oxidation, ATR-FTIR spectroscopy was performed on PP12K, as shown in Figure 4. Presently relevant are the ones for the untreated pellet and the PP scrapings heated at 140 °C in air for 24 h. As shown in the figure, the emergence of a peak at 1715 cm<sup>−1</sup> and a shoulder at 1770 cm<sup>−1</sup> indicates the formation of carbonyl containing



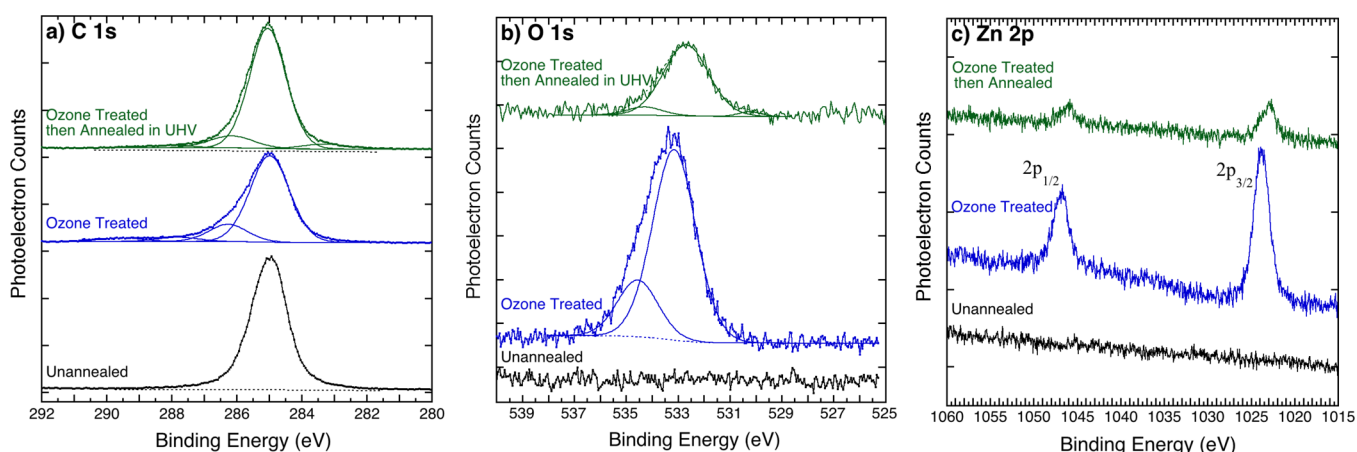
**Figure 4.** ATR-FTIR of unannealed, annealed-in-air at 140 °C, and ozone-treated PP12k. The peaks at 1708 and 1770 cm<sup>−1</sup> are indicative of the formation of carbonyl (e.g., ketones) and carboxylic acid groups.

functional groups on the PP, including ketones, carboxylic acids, and esters.<sup>20</sup>

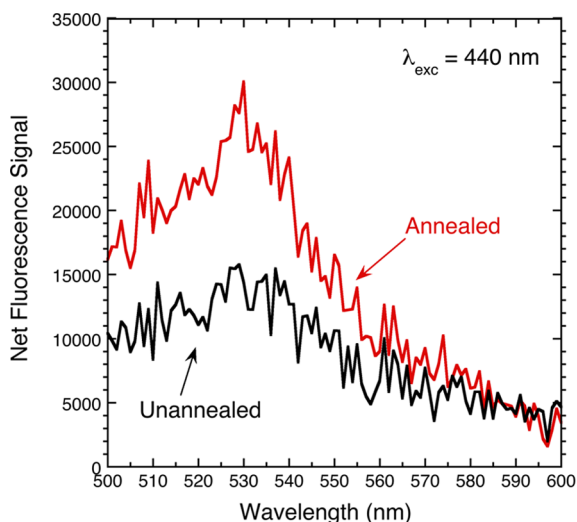
Ozone treatment is particularly interesting. These experiments simply involved exposing the films to ozone gas; no heating was performed. In the case of the PP12k films, the percentages of Zn and O increase to 0.66 and 11%, respectively. For the PP250k film, the corresponding values are 0.050 and 9.7%. This experiment shows that surface segregation of ZnO NPs occurs for the low molecular weight film but to a significantly less extent for the high molecular weight one, even though both samples are similarly oxidized. This likely results from the reduced mobility of the longer polymer chains in the PP250k sample, which prevents ZnO NPs from reaching the surface.

To determine whether the surface segregation of ZnO NPs is reversible, the following experiment was performed. The 5 wt % ZnO/PP12k film sample that had been exposed to ozone and that displayed a strong Zn2p signal was heated in ultrahigh vacuum at 140 °C for 24 h. After it cooled to room temperature, XPS was remeasured. As shown in Figure 5, the Zn2p XPS peak decreased due to annealing. The O1s peak intensity and C1s components indicative of oxidation also decrease. These results indicate that the desorption of the polar functional groups that arise from ozone exposure leads to the disappearance of most of the ZnO NPs from the surface as they diffuse back into the bulk.

**ZnO NP Surface Activity.** To confirm that surface-segregated ZnO NPs are available for surface chemical reactions (e.g., adsorption and catalysis), a fluorescent dye (FMN) that selectively binds to ZnO NPs was employed. Phosphonates are known to link covalently to ZnO nanoparticles.<sup>21</sup> FMN is a phosphonate terminated fluorophore that may be excited at 470 nm, with a maximum emission at ca. 550 nm.<sup>22</sup> The situation is complicated slightly by the fact that ZnO NPs are also fluorescent, with absorbance in the UV, at ca. 340 nm.<sup>23</sup> Therefore, 440 nm excitation, which is the closest accessible wavelength near 470 nm with the confocal laser fluorescent microscope, was chosen. Figure 6 shows (corrected) emission spectra from a ZnO/PP250K film that was annealed at 160 °C for 24 h and one that was not annealed; both were immersed in aqueous 1.0 mM FMN and then copiously rinsed with water. Because some nonspecific



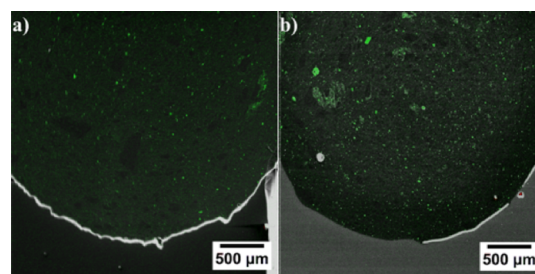
**Figure 5.** Al K $\alpha$  XPS of the (a) C1s, (b) O1s, and (c) Zn2p regions of a spin-coated film of ZnO/PP12k on a silicon wafer that was exposed to ozone to induce ZnO migration to the surface. These are the same data as in Figure 2. However, the sample was then annealed at 140 °C for 24 h in UHV to remove some of the oxidized components on the surface (e.g., carboxylic acid groups). The disappearance of the Zn2p peaks indicates the reversibility of the surface segregation process. Peak fits are included for spectra that have more than one significant component.



**Figure 6.** Fluorescence spectra of ZnO/PP250k spin-coated films after immersion in ethanolic FMN solution, rinsing, and drying. One of the films was heated in air for 24 h at 160 °C ("annealed"). The other was immersed without annealing. Both are difference spectra, with the spectrum corresponding to a PP250k film (immersed in FMN, rinsed, and dried) subtracted from them. The excitation wavelength was 440 nm.

adsorption of FMN occurs on all the surfaces, the corrected emission spectra are the difference between the sample of interest and PP250k (without ZnO) that was immersed in the FMN solution. The greater emission intensity at 530 nm is consistent with the conclusion that ZnO NPs migrate to the surface after heating in air.

To gain further insight into the surface segregation phenomenon, cross-sectional FE-SEM, combined with EDS mapping, was performed on a fiber made from PP12k blended with 3 wt % ZnO NPs. It should be noted that these fibers lack mechanical strength and flexibility due to the low molecular weight, but the lower molecular weight PP fiber was chosen because it was thought that it would have the best chance of exhibiting detectable surface segregation from the bulk. Figure 7 depicts cross-section images of the unannealed and 140 °C (in air) annealed fibers. The green dots correspond to ZnO NPs, as determined by EDS of Zn. As discussed in the previous

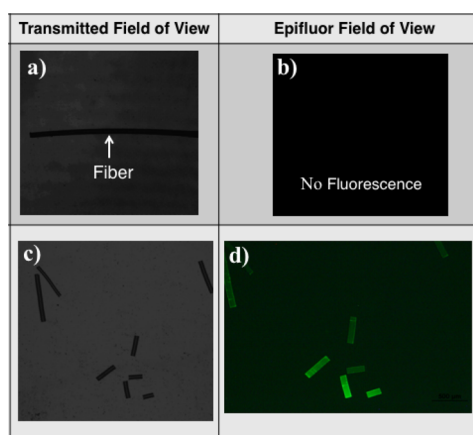


**Figure 7.** FE-SEM/EDS cross-section images of a ZnO/PP12k fiber (a) before and (b) after annealing at 140 °C. The green dots are ZnO nanoparticles. The similarity of the two images indicates that ZnO NPs do not diffuse and deplete the bulk of the fiber of particles.

section, this annealing procedure results in the surface segregation of the ZnO particles in PP12k films. However, it is observed that both images contain similar numbers of ZnO nanoparticles uniformly dispersed within the bulk of the fiber. These data indicate that while diffusion of the NPs to the surface takes place, it only occurs in the near-surface region.

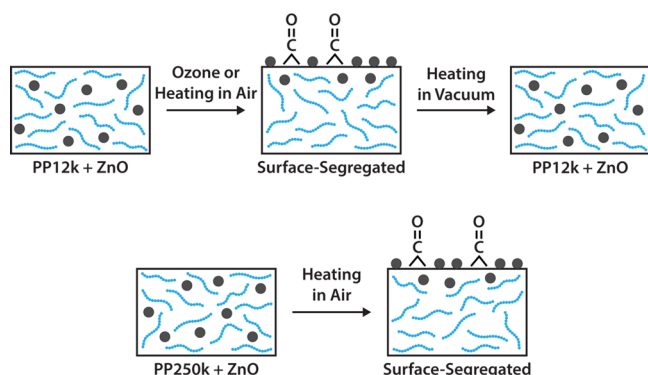
To confirm the phenomenon and that ZnO particles have indeed surface segregated and that they are available for adsorption/reaction, a fiber extruded from PP250k blended with 5 wt % ZnO NPs was annealed in air at 160 °C. The fiber became brittle following this procedure and fractured into smaller pieces that were immersed in the 1.0 mM FMN solution, rinsed in water, and dried. Transmission and confocal fluorescence microscopy images of the fiber pieces are depicted in Figure 8. It was confirmed that a similarly annealed PP250k fiber subsequently immersed in the FMN solution did not fluoresce. The green fluorescence indicates that ZnO NPs have surface segregated and bonded to the fluorescent FMN dye.

**Surface Segregation Mechanism.** One possible rationalization for the observed results is that oxidation induces chain mobility and uncovering of the MONPs. This would be a logical explanation if heating, which results in increased chain mobility, was necessary in all cases for the phenomenon to be observed. However, because ozone exposure without heating leads to the emergence of ZnO nanoparticles on the surface of PP12k films and fibers, this explanation seems unlikely. Instead, we hypothesize that the diffusion of the nanoparticles to the surface of the fibers and films is responsible for the



**Figure 8.** Transmission light and confocal fluorescence microscope images of 3 wt % ZnO PP250k fibers after immersion in an aqueous FMN solution and rinsing. Images a and b are for an unannealed fiber, and images c and d correspond to a fiber that was annealed in air at 160 °C for 24 h. The annealed fiber is in smaller pieces because it became brittle due to annealing. The green fluorescence indicates that FMN has adsorbed on the annealed fiber surface.

observed results. The proposed process is summarized in Figure 9.



**Figure 9.** Schematic of the ZnO NP surface segregation process for PP12k and PP250k fibers and films. In the case of PP12k, either ozone treatment or heating in an oxygen-containing atmosphere leads to ZnO NP surface segregation. However, for PP250k, ozone exposure does not result in surface segregation due to the limited chain mobility. For PP12k, surface segregation may be reversed by heating the ZnO surface-segregated sample in UHV to desorb polar groups.

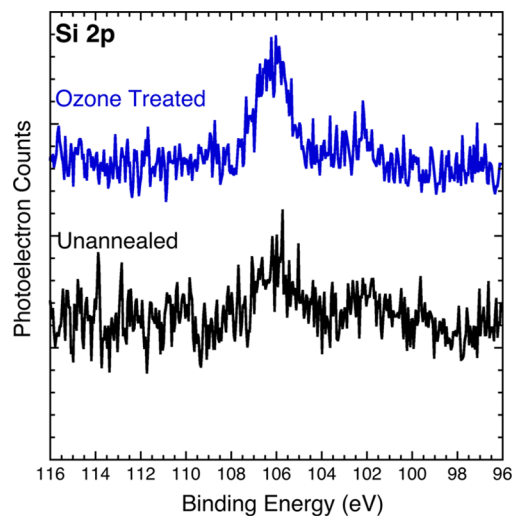
Surface and bulk segregation of nanoparticles in a polymer melt has previously been studied. For a system with a negligible enthalpic interaction between polymer and particles, entropy effects dominate. For example, Krishnan et al.<sup>24</sup> studied the migration behavior of polystyrene nanoparticles uniformly distributed in linear polystyrene. The authors found that annealing the system above the glass transition temperature led to the segregation of the particles to the silanized silicon oxide substrate, with the process driven by the increased entropy of the polymer chains. This increase in polymer chain entropy was enough to overcome the loss of a degree of freedom of translational entropy of the segregated particles. In contrast, oleic acid-capped CdSe NPs embedded in polystyrene migrated to the air interface due to the low surface free energy of the oleic acid layer on the NPs. Nowicki<sup>25</sup> has

discussed the conformation entropy of long polymer chains in the presence of both nonadsorbing and attached colloidal particles. The entropy-driven effect of migration of the particle toward the periphery of the polymer coil may be overcome by the enthalpic effect of particle adsorption, but the competition between the two effects depends on the particle size and shape, polymer chain length, and adsorption energy of the particle to the polymer.

In the present study, surface segregation does not occur when ZnO NP-containing PP films are heated in the absence of oxygen. It only occurs if polar groups, arising from surface oxidation, are present. If entropic effects dominated, it would occur in both cases due to the same gain of entropy of the polymer chains. The fact that it only occurs when the surface is oxidized strongly suggests that enthalpic effects are responsible for the surface segregation.

The ZnO NPs used in this study are essentially fully hydroxylated.<sup>23</sup> In a previous investigation,<sup>15</sup> we compared adhesive forces between a ZnO particle attached to a cantilever and various PP films with atomic force microscopy. It was found that the pull-off force was significantly greater for oxidized PP compared to an untreated film. The result may be explained by dipole–dipole, hydrogen bonding, or possibly covalent bonding between the hydroxyl groups of the ZnO and the carbonyl-containing functional groups on the PP surface (e.g., carboxylic acid groups). It is therefore postulated that the enthalpic attraction between surface-segregated ZnO NPs and polar functional groups on oxidized PP fibers and films drives the surface segregation.

Some preliminary work has been performed to investigate whether other MONPs exhibit a similar behavior when embedded in PP. Spin-coated PP films containing ca. 20 nm silica nanoparticles were prepared on MgO(100) substrates. These substrates were used instead of silicon wafers to avoid the possibility of XPS confusing the substrate signal with that of the NP. As shown in Figure 10, ozone exposure of a PP12k film containing 5 wt % SiO<sub>x</sub> particles causes a dramatic increase in the Si2p XPS signal, although some Si2p signal is present even before ozone treatment. However, ozone exposure causes the % Si (on a carbon basis) to increase



**Figure 10.** Al K $\alpha$  XPS of the Si2p region of an untreated 5 wt % SiO<sub>x</sub> NP in the PP film spin-coated on MgO(100) before and after exposure to ozone.

from 1.1 to 1.5%. This result demonstrates that other MONPs will likely exhibit a similar surface segregation behavior as that of the ZnO NPs studied in detail in this paper.

## CONCLUSIONS

The present study shows that ZnO NPs embedded in either PP films or fibers migrate to the air–polymer interface when annealed in air or, to a lesser extent, when annealed in a poor vacuum. However, annealing in UHV does not lead to this surface segregation. Exposure to ozone gas at room temperature also causes surface segregation but only for the low molecular weight PP film, presumably because of the limited chain mobility of the higher molecular weight chains, which apparently block the NPs from reaching the surface. Heating PP films in oxygen-containing atmospheres or ozone treatment causes the formation of ketone and carboxylic acid groups on the polymer surface. The adsorption of a fluorescent, phosphonated dye on the surface-segregated particles indicates that the ZnO NPs are available for surface reactions. Heating the PP12k film containing surface-segregated NPs in UHV to remove much of the ketone/carboxylic acid groups causes the NPs to return to below the surface. As a whole, these results demonstrate that surface segregation is enthalpically driven by the attraction of the particles to the oxygen-containing functional groups. Preliminary experiments on silica NPs indicate that they exhibit a similar surface segregation behavior. Further work is planned to investigate other MONPs, including ones having different shapes and sizes. The results in this paper indicate that it may be possible to fabricate polymeric fibers whose surfaces are functionalized with metal oxide nanoparticles simply by melt extruding a blended mixture of the polymer and nanoparticles and then oxidizing the surface of the fibers. However, the method has potential limitations compared to established state-of-the-art techniques. These include fiber damage, due to chain scission, that may affect the mechanical properties of the fibers, and lower surface concentration of the nanoparticles compared to methods such as ALD. However, if the process is optimized, there may be niche applications for the methodology.

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## Notes

The authors declare no competing financial interest.

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